

RTL Verification and FPGA Implementation of 4x4 Vedic Multiplier

Mohd Esa , Konasagar Achyut and Chandrajeet Singh

Abstract. The objective of this paper is to study 4x4 Vedic multiplier. Multiplication is an important fundamental function in arithmetic operations. Vedic multiplier using Urdhva-Tiryagbyam sutra is predominant in performance evaluation of parameters such as power, area & delay. This paper presents design, verification and FPGA implementation of Vedic multiplier. Verification is carried out in Questa Sim 10.4e using System Verilog HVL and design is carried out in Xilinx ISE Design Suite 14.7 using Verilog HDL environment.

Keywords: Vedic Multiplier, HVL, HDL, RTL, FPGA.

1 Introduction

Binary number system uses only zero's and one's. Just like decimal system, binary possesses every arithmetic operation. A binary multiplier is any such electronic circuit used in digital electronics to multiply two binary numbers [1]. Unlike the decimal base ten, binary multiplication is done in binary base two. The concept of Vedic multiplier has been acquired from the Vedic mathematics in which there are several methods to operate with the number systems. Urdhva-Tiryagbyam sutra is one among those Vedic methods which helps to follow general formula applicable to all cases in multiplication. The meaning of Urdhva-Tiryagbyam is vertically and crosswise.

This paper presents the design and verification of 4x4 bit Vedic multiplier using HDL and HVL respectively, which helps us to justify that design is working without any bugs or errors. This paper also presents implementation of Vedic multiplier in Field Programming Gate Array. FPGAs are semiconductor devices that are based around a matrix of configurable logic blocks connected via programmable interconnects. FPGAs can be reprogrammed to desired application or functionality requirements after manufacturing.

2 Ripple Carry Adder

As the name of the circuit itself represents that the carry is rippled to a succeeding part. The combination of full adders whose carry output is propagated as a

carry input to the succeeding full adder is called as a Ripple Carry Adder. The addition of two binary numbers in parallel implies that all the bits of addend and augend are available for computation at the same time [2].

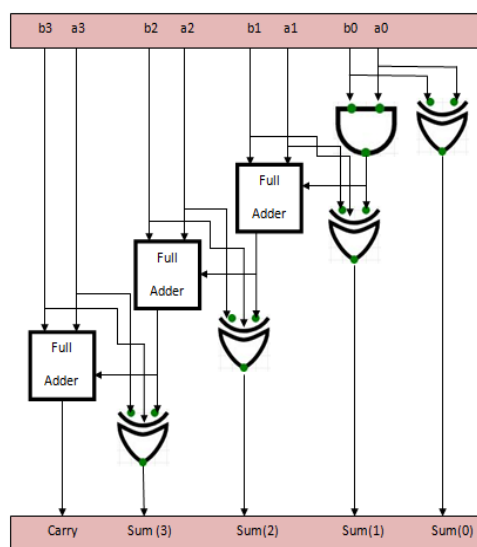


Fig.1. Ripple Carry Adder logic circuit

Table 1. Ripple Carry Adder truth table

Inputs		Outputs	
$[A_3-A_0]$	$[B_3-B_0]$	Sum	Carry
0000	0001	0001	0
0010	0010	0100	0
0011	0101	1000	0
1011	1010	0101	1
1101	1101	1010	1
1110	0111	0101	1
1000	1111	0111	1
1110	1001	0111	1
0101	0011	1000	0
1100	0110	0010	1
-	-	-	-
-	-	-	-
1111	1111	1110	1

In this work, three ripple carry adders are used in which the first bits from addend and augend binary numbers are computed by half adders and rest with full adders to form a complete ripple addition. The truth table for various possible combinations of ripple carry adder is shown table 1.

3 Urdhva Triyagbhyam Sutra

Mathematics is mother of all sciences, it is full of mysteries and magic. Ancient Indians were able to understand these mysteries and developed simple keys to solve them. The ancient system of Vedic maths was introduced by Swami Bharati Krishna Tirthaji. His work includes various methods of calculations and this in turn Indians named it as Vedic mathematics [3].

This paper presents multiplication operation done over two 4-bit binary numbers through the Urdhva-Triyagbhyam technique. As usual the multiplication has a multiplier and a multiplicand, but the method of multiplication is done by multiplying first two bits of multiplicand and multiplier followed by cross multiplication of side numbers of multiplicand with numbers in multiplier, finally the last digits are multiplied parallelly to complete the multiplication process. This method can be observed as shown in the fig.2 for two bit binary numbers. To perform this in digital logic the block diagram of two bit Vedic multiplier is shown in fig.3.

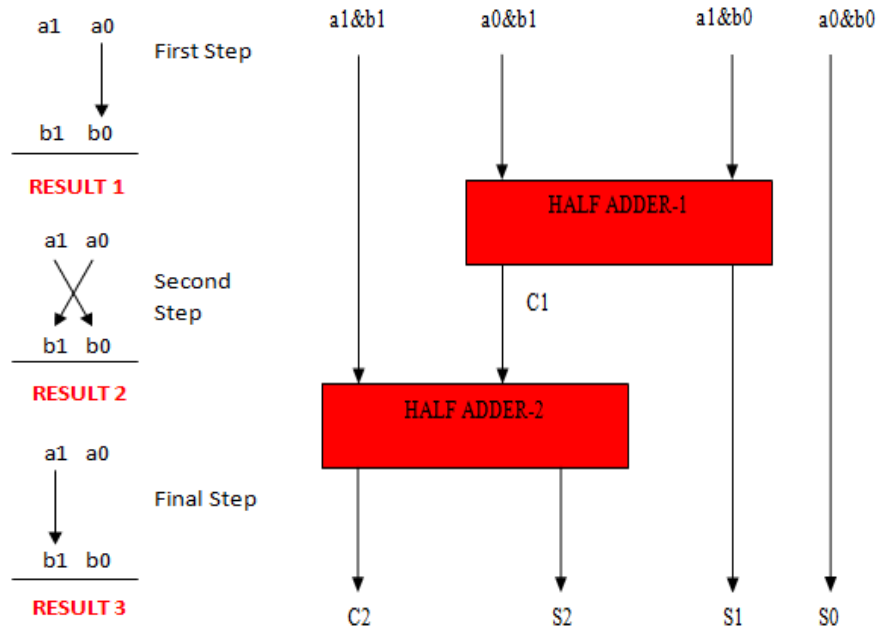


Fig.2. 2x2 Multiplication Flow

Fig.3. Logic Diagram of 2x2 Vedic Multiplier

Now, since our paper presents the work of four bit binary multiplication we have formed a circuit which follows the same method as two bit binary multiplication. The structure of four bit binary multiplication its dot diagram is shown in fig.4 and fig.5 respectively.

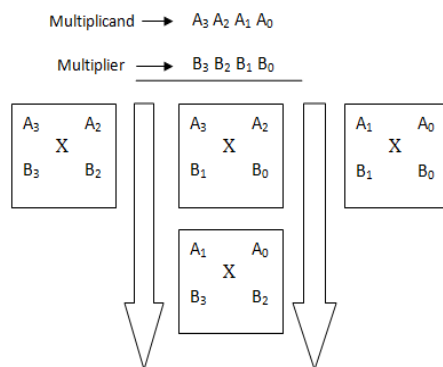


Fig.4. Mathematical structure of 4x4 Vedic multiplier

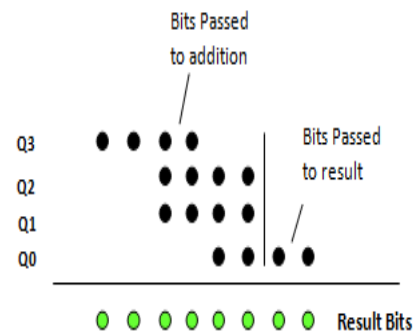


Fig.5. Dot diagram of 4x4 Vedic multiplier

By the help of two bit Vedic multiplier we are able to construct four bit Vedic multiplier which includes ripple carry adders to add the bits parallelly.

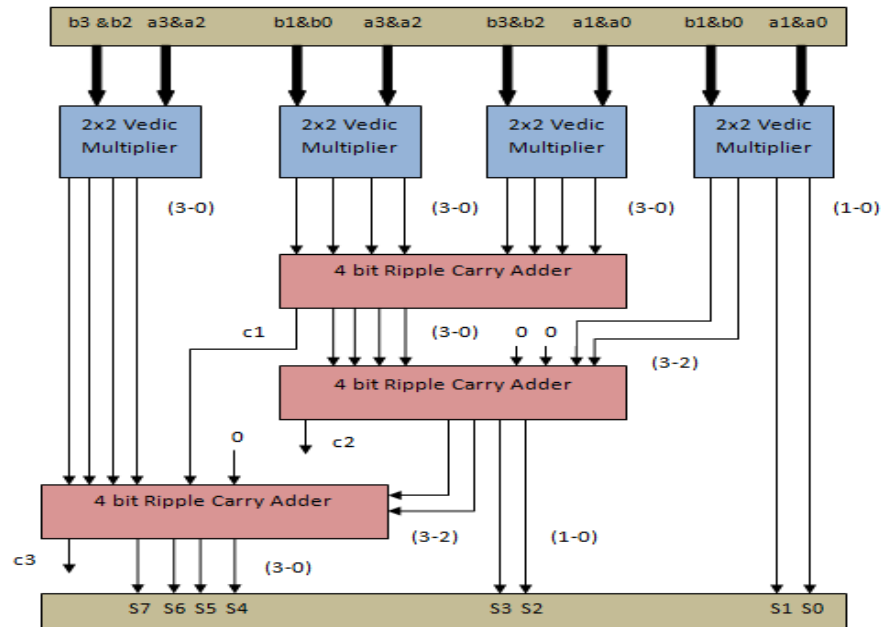


Fig.6. Architecture of 4x4 Vedic Multiplier

4 Simulation results

Simulation provides an effective way to investigate the design which can be easily rectified by means of EDA tools. Fig.7 shows simulation waveforms of inputs as well as output using Verilog HDL & fig.8 shows RTL schematic of Vedic Multiplier.

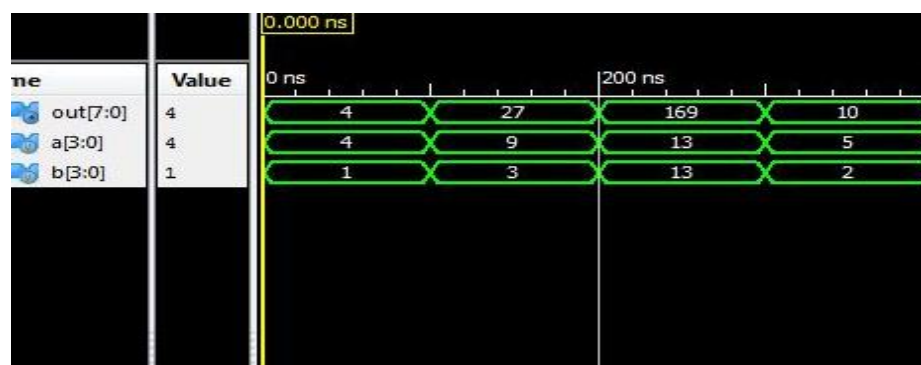


Fig. 7. Simulation output of 4x4 Vedic Multiplier (Using Verilog)

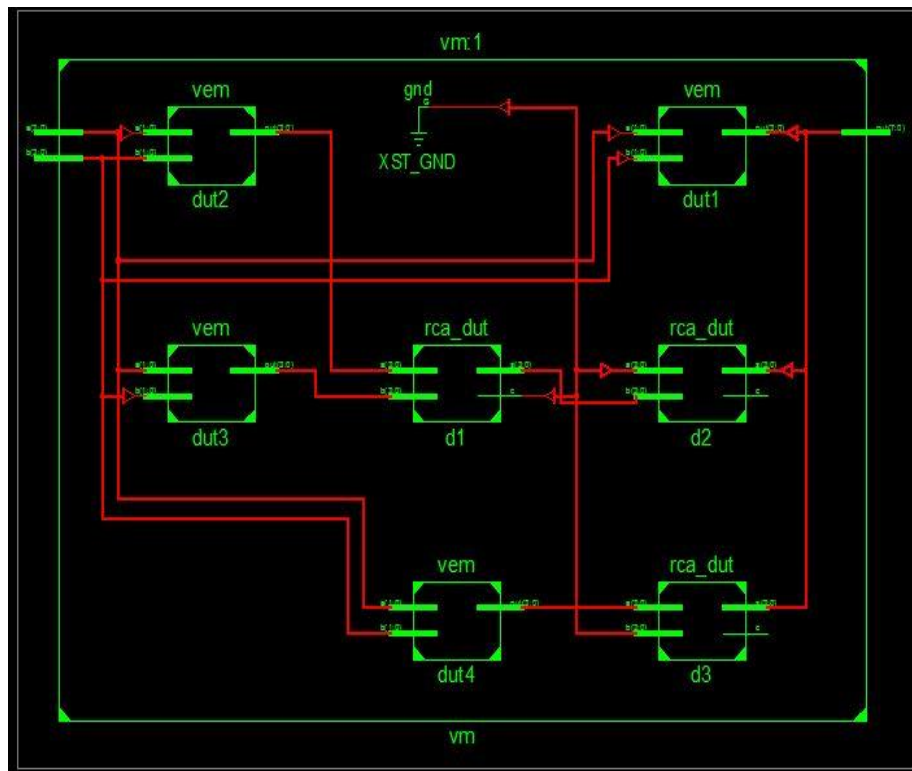


Fig.8. RTL schematic of Vedic Multiplier

Verification is carried out using object oriented programming concept in System Verilog hardware verification language and design is justified error free using functional coverage concept. Fig.10 shows the 100% functional coverage report for different random inputs.

	Msgs				
+ /vm_verif/vif/a	4'h2	4'h3	4'h8	4'h0	
+ /vm_verif/vif/b	4'h1	4'hd	4'h7	4'h4	4'h8
+ /vm_verif/vif/out	8'h02	8'h27	8'h38	8'h00	

Fig.9. Verification of Vedic Multiplier by generating random inputs (Using System Verilog)

Number of tests run:	1
Passed:	1
Warning:	0
Error:	0
Fatal:	0

[List of tests included in report...](#)
[List of global attributes included in report...](#)
[List of Design Units included in report...](#)

Coverage Summary by Structure:			Coverage Summary by Type:						
Design Scope *	Hits % *	Coverage % *	Total Coverage:						
vm_top_sv_unit	100.00%	100.00%	Coverage Type *	Bins *	Hits *	Misses *	Weight *	% Hit *	Coverage *
vm_gen/gen_run#ublk=166586548#15	100.00%	100.00%	Covergroups	3	3	0	1	100.00%	100.00%
coverage	100.00%	100.00%	Assertions	1	1	0	1	100.00%	100.00%

Fig.10.Coverage report of Vedic Multiplier using SV environment

FPGA board results for two binary numbers “0101” and “0111” which equals “00100011” can be observed in fig.11 where pins from LD0 to LD7 indicates eight bits, if any of LD pin is seems to be glowing it represents logic ‘1’ else logic ‘0’.

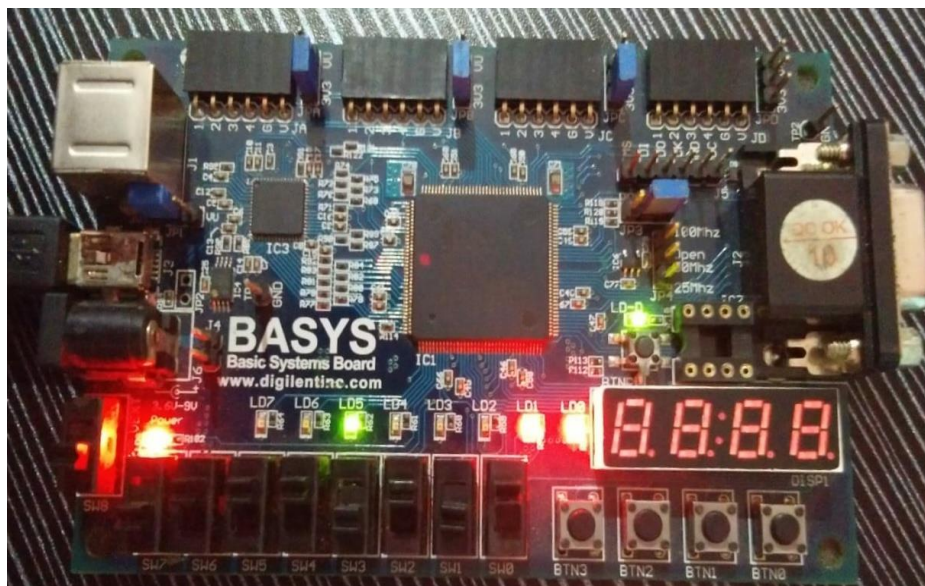


Fig.11.FPGA implementation of 4x4 Vedic multiplier

5 Conclusion

In this paper, we have studied about multiplication of two 4-bit binary numbers using Vedic multiplier. Vedic multiplier is designed by means of HDL. After successful simulation of the required design it has been verified for its functionality using HVL with functional coverage. From fig.7 and fig.9 it concludes that design is meeting the expected result covering all the verification scenarios explicitly. Also Vedic multiplier is successfully implemented in FPGA.

References

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Profiles



Mohd Esa completed M.E. (Power Electronics systems) from Muffakham Jah College of Engineering and Technology, Banjarahills, Hyderabad in 2018. He received his B.E degree from Matrusri Engineering College, Hyderabad in 2015. He was awarded gold medal twice for standing first in B.E. III/IV and B.E. IV/IV from Matrusri Engineering College, Sayeedabad, Hyderabad. He has published 11 research papers in various journals and conferences. His research of interests includes Multi level inverters and Multipliers. He is trained VLSI Design Engineer.



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Chandrajeet Singh is trained in VLSI Front End RTL Design Engineer. He received his Bachelor of Technology Degree in 2015 specialized in Electronics and Communication Engineering, from BITS College, Bhopal (Affiliated to RGPV, Bhopal). His research of interest includes Encryption Standards, Double Hash algorithm and Polar codes.